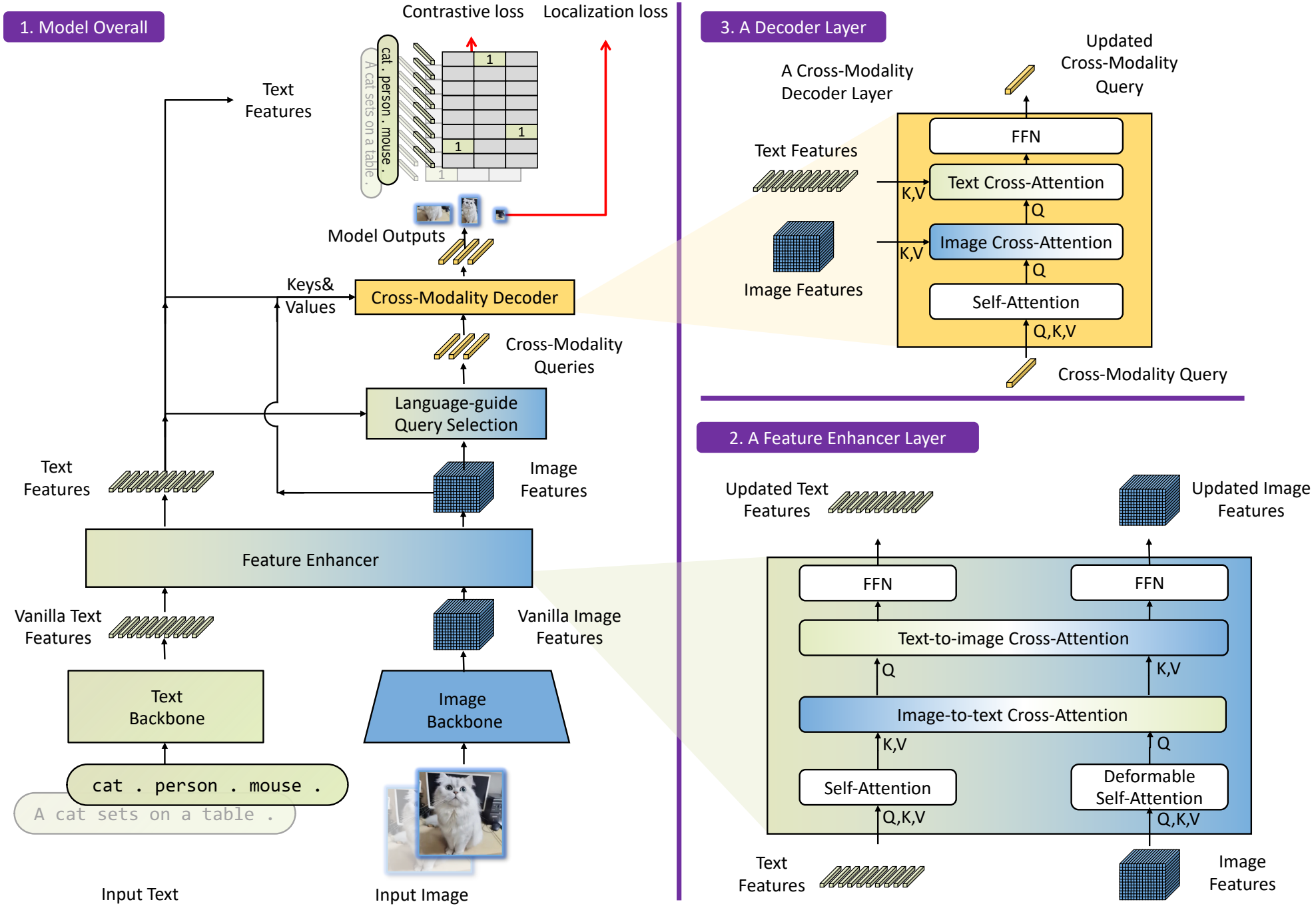

1. Model Overall
Contrastive loss
Localization loss
cat . person . mouse .
A cat sets on a table .
Text Features
Model Outputs
Keys& Values
Cross-Modality Decoder
Cross-Modality Queries
Language-guide Query Selection
Text Features
Image Features
Feature Enhancer
Vanilla Text Features
Vanilla Image Features
Text Backbone
Image Backbone
cat . person . mouse .
A cat sets on a table .
Input Text
Input Image
3. A Decoder Layer
A Cross-Modality Decoder Layer
Updated Cross-Modality Query
Text Features
Image Features
FFN
Text Cross-Attention
Image Cross-Attention
Self-Attention
K,V
Q
Q,K,V
Cross-Modality Query
2. A Feature Enhancer Layer
Updated Text Features
Updated Image Features
FFN
Text-to-image Cross-Attention
Image-to-text Cross-Attention
Self-Attention
Deformable Self-Attention
Q
K,V
Q,K,V
Text Features
Image Features